

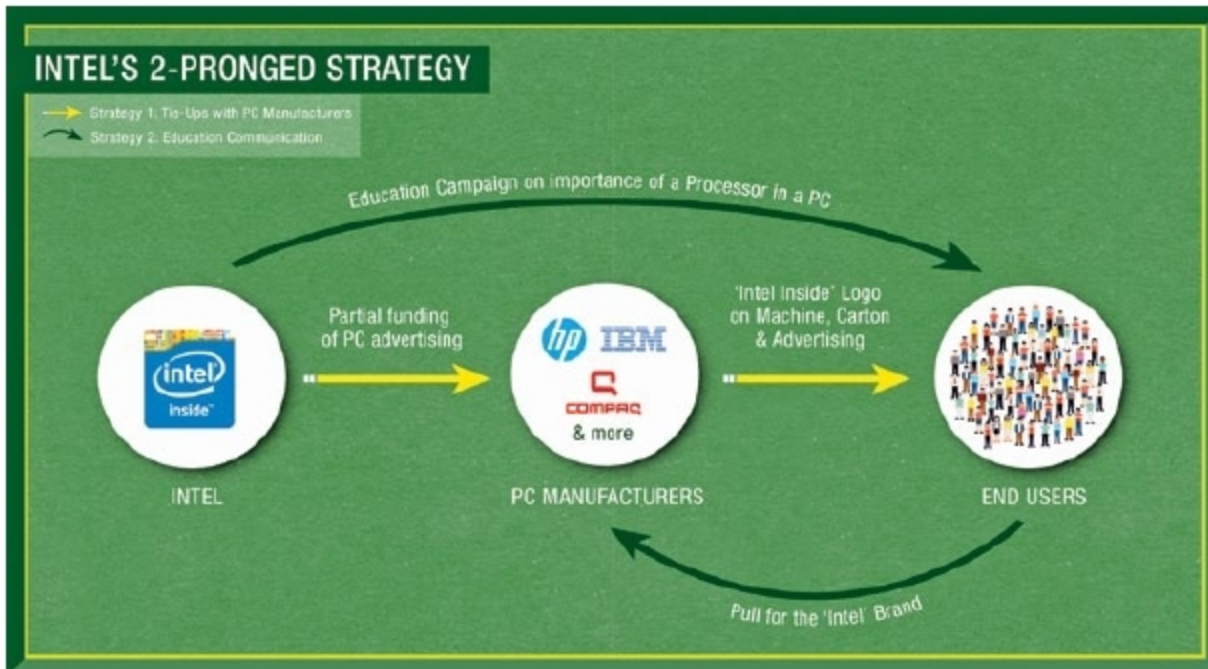


Fig. 4: Intel's two-pronged marketing strategy



Fig. 5: Ray Dolby, inventor of noise reduction system known as Dolby NR

he founded the company. The first technology he came up with was a first-in-the-world one. He figured out how to reduce the hiss that happens while recording sound. This he called noise reduction (NR).

That was the beginning of an incredible journey. Dolby first got sound studios in England to adopt NR. Soon, US sound studios followed suite. He rode the whole wave of cassette tapes and tape players. He moved into providing NR sound for movie theatres.

At the very beginning, Dolby decided it would manufacture professional products only and license technologies appropriate for consumers. But the big inflection point came when *Star Wars* used Dolby to provide sound for the film. Even though state-of-the-art sound was being used in albums and concerts, in movies it was a commoditised category. Ste-

phen Katz, sound consultant from Dolby who worked on *Star Wars*, said, "Sound was a very important component for *Star Wars*, and it was a component that Hollywood virtually ignored. People used to tell me, 'Nobody listens to the sound.'" Well, not until George Lucas decided to change it. It is his vision that Dolby brought to life.

None of the movie theatres were fitted with Dolby sound, but Twentieth Century Fox insisted that if they wanted the 70mm print (vs the 35mm), they needed to install Dolby. And because very few movie theatres had made that shift, initially *Star Wars* released with only 40 prints as against the usual 800 prints. And, only three prints had Dolby sound.

The film created such magic that things took off from there. Movie theatres clamored to have Dolby sound. Without *Star Wars*, Dolby would not be where it is now. Also, without Dolby, *Star Wars* would not be as magical, either. Sound added a lot to the film. So much so that the industry henceforth looked at sound differently. It was no longer a commoditised product.

Dolby's journey did have a hiccup when it did not realise the impact of digital audio on cinemas. Sound until now, although of excellent quality, was analogue. And, an entrepreneur, Terry Beard along with Steven Spielberg, Universal Studios and other investors created Digital Theatre Systems (DTS). Dolby lost out and took a few years to gain back the leadership,

which it determinedly did.

Dolby's business focus has been innovation and patent protection. It has made most of its money through licensing. As of 2018, it still makes 90 per cent of its revenue through licensing, and 10 per cent through products and services.

Now, Dolby has proliferated every aspect of our lives. The movie theatre product is Dolby Atmos, which it has moved beyond audio into video by adding Dolby Vision.

Its home theatre product is Dolby Digital Plus. Dolby is working with companies to provide its products on smartphones, TVs, games, tablets, streaming services, headphones, movie theatres, sound studios and more.

Takeaway. Dolby is a company that has innovated consistently to improve its own products. It has worked to make itself redundant, so no one else can. This focus on innovation has meant that it has rarely needed to rely on buying technology from outside the company. The decision to focus on top-end professionals and leave the rest to licensing ensured a continuous inflow of funds, so much so that it did not feel the need to go public until



Fig. 6: *Star Wars* used Dolby to provide sound for the film